ModelIt!

Teacher's Guide

# The Science of Sleep: Why Teens Can't Wake Up

Modeling Circadian Rhythms, Melatonin, and Sleep Architecture

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 12L1 |
| **NGSS Standard** | HS-LS1-3 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

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# Table of Contents

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1 NGSS Standards Unpacking & CAST Alignment ..... 3

2 Scoring Rubric ..... 4

3 Background Content Knowledge ..... 5

4 The LEVER Framework ..... 6

5 Slide-by-Slide Facilitation Guide ..... 7

6 Answer Key & Expected Responses ..... 9

7 Collaborative Learning Protocols ..... 10

8 Differentiation & Extensions ..... 11

9 Common Misconceptions ..... 12

# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: HS-LS1-3**

Plan and conduct an investigation to provide evidence that feedback mechanisms maintain homeostasis.

## Three-Dimensional Learning

| | |
|---|---|
| **Science Practice** | **Developing and Using Models** — Students develop computational models to predict how light exposure, circadian phase, and sleep pressure interact to determine sleep quality and cognitive performance. |
| **Disciplinary Core Idea** | **LS1.A Structure and Function** — Feedback mechanisms including the circadian clock and sleep homeostasis maintain the body's sleep-wake cycle, and disruptions to these mechanisms impair cognitive and physiological function. |
| **Crosscutting Concept** | **Stability and Change** — Students analyze how the circadian system maintains stable sleep-wake cycles through feedback mechanisms and how external disruptions (light, schedules) shift the system away from equilibrium. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A student uses their phone in bed until 11:30 PM and must wake at 6:30 AM for school. Based on the sleep model, which component is most directly disrupted first, and what cascade of effects follows?

**CONSTRUCTED RESPONSE:**

Using your model, explain why the American Academy of Pediatrics recommends high school start times no earlier than 8:30 AM. Reference the circadian phase shift during adolescence and its effects on at least three components.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Sleep is not a passive state — it's an active, highly structured process essential for memory consolidation, emotional regulation, immune function, and physical growth. During adolescence, a biological shift in the circadian clock creates a perfect storm: the brain wants to sleep later, but society demands earlier wake times, resulting in chronic sleep deprivation that affects millions of students.

## The Two-Process Model of Sleep

Sleep is regulated by two independent systems: Process C (circadian rhythm) and Process S (sleep homeostasis). Process C is controlled by the suprachiasmatic nucleus, a cluster of ~20,000 neurons in the hypothalamus that tracks the 24-hour day using light as the primary cue. Process S is driven by the accumulation of adenosine in the brain during waking hours. When both systems align — high sleep pressure meeting circadian sleep phase — you fall asleep quickly and sleep well.

## The Adolescent Circadian Shift

During puberty, the circadian clock shifts approximately 1-3 hours later due to changes in melatonin secretion timing and sensitivity to light. This is not a choice or a habit — it's a fundamental change in brain biology. Melatonin onset, which occurs around 8-9 PM in pre-pubescent children, shifts to 10-11 PM in adolescents. This means a teenager physically cannot fall asleep as early as a child or adult, even with perfect sleep hygiene.

## Blue Light and Melatonin Suppression

The photoreceptors responsible for setting the circadian clock are particularly sensitive to blue-wavelength light (460-480nm). Electronic screens emit significant blue light, and exposure within 2 hours of bedtime has been shown to suppress melatonin production by up to 50%, delay sleep onset by 30-90 minutes, and reduce REM sleep. For adolescents with already-delayed circadian phases, screen time before bed compounds the biological shift with a behavioral one.

## Sleep Deprivation and Academic Performance

The CDC reports that 73% of high school students get less than the recommended 8-10 hours of sleep. Research shows that each hour of sleep lost below optimal is associated with a 0.07 GPA decrease, increased absenteeism, and higher rates of anxiety and depression. Schools that have adopted later start times (8:30 AM or later) report 4-5% increases in standardized test scores, reduced car accidents among teen drivers, and improved mental health outcomes.

# 4 The LEVER Framework

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### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**
Students identify light exposure, circadian phase, melatonin level, sleep pressure, and cognitive performance as the key components of the teen sleep system.

**E** **Establish Relationships**
Students determine that light exposure sets circadian phase, darkness triggers melatonin, waking hours build sleep pressure, and alignment of these factors determines cognitive performance.

**V** **Visualize & Model**
Students build a computational model showing how the circadian system, melatonin, and sleep pressure interact to determine when teens naturally sleep and how well they function.

**E** **Evaluate Outcomes**
Students run natural schedule, blue light disruption, and early school start scenarios to test predictions about teen sleep biology versus societal demands.

**R** **Revise & Extend**
Students add caffeine, REM proportion, or cortisol to explore how common teen behaviors and stress interact with the basic sleep system.

### Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"Today we're investigating something that affects every single one of you. Modeling Circadian Rhythms and Sleep Architecture."*

**Watch For**

Show cover image. Ask: What do you already know about this topic? Quick share-out.

## Slide 2: Learning Objectives

3 min

**Say:** *"Here's what you'll be able to do by the end of today's lesson."*

**Watch For**

Have students read objectives aloud. Pre-teach vocabulary with quick visual aids.

## Slide 3: Big Question

3 min

**Say:** *"This is our driving question. By the end of class, you'll be able to answer it with evidence from your model."*

**Watch For**

Quick-write: Students write their initial hypothesis. Save to compare later.

## Slide 4: LEVER Framework

2 min

**Say:** *"We're going to build a computational model to investigate this system. Here are our five steps."*

**Watch For**

Preview each LEVER step. Emphasize that modeling reveals hidden patterns.

## Slide 5: Activity 1: Components

8 min

**Say:** *"What are the key parts of this system? Which ones come from outside, and which respond inside?"*

**Watch For**

Guide sorting of external vs. internal components. Use think-pair-share for reasoning.

## Slide 6: Activity 2: Connections

8 min

**Say:** *"During puberty, melatonin doesn't rise until 11 PM. School starts at 7:30 AM. What happens?"*

**Watch For**

Students draw arrows showing +/- relationships. Discuss feedback loops if present.

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"Now let's test our model. Run each scenario and record what happens."*

**Watch For**

Students run all scenarios, record predictions vs. observations. Discuss surprises.

## Slide 8: Discoveries

5 min

**Say:** *"So what did our model reveal? Teens can't wake up because their brains are wired to sleep later during puberty...."*

**Watch For**

Class discussion of discoveries. Compare initial hypotheses to model evidence.

## Slide 9: STEM Challenge

12 min

**Say:** *"Now apply what you've learned. Design an optimal school schedule maximizing cognitive performance...."*

**Watch For**

Groups design solutions using model data. Present and evaluate designs.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Light Exposure
- **INTERNAL:** Circadian Phase, Melatonin Level, Sleep Pressure, Cognitive Performance

## Relationships

- **Light Exposure to Circadian Phase:** DETERMINES — Light, especially blue wavelength, is the primary signal that sets the timing of the circadian clock. Morning light advances the clock (earlier sleep); evening light delays it (later sleep).
- **Circadian Phase to Melatonin Level:** DETERMINES — The circadian clock triggers melatonin production as the sleep phase approaches. In teens, this trigger occurs about 2 hours later than in adults.
- **Melatonin Level to Sleep Pressure:** POSITIVE (+) — Rising melatonin reinforces the biological drive to sleep, working with accumulated adenosine to promote sleep onset.
- **Sleep Pressure to Cognitive Performance:** NEGATIVE (-) — Unresolved sleep pressure from insufficient or poorly timed sleep directly impairs memory, attention, and executive function.

## Key Discoveries

- The adolescent circadian shift is a biological reality — melatonin onset occurs 1-2 hours later in teens, making early wake times fight fundamental biology
- Blue light from screens at night directly suppresses melatonin production, delaying sleep onset by 30-90 minutes even when the person feels tired
- Sleep debt is cumulative — each night of insufficient sleep compounds cognitive impairment, and weekend catch-up only partially restores function
- Cognitive performance is best when sleep timing aligns with circadian phase — it's not just total hours but WHEN those hours occur that matters

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
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| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Teenagers who stay up late are just being lazy or irresponsible" | The adolescent circadian shift is a well-documented biological phenomenon driven by changes in melatonin secretion timing during puberty. Telling a teenager to go to bed at 9 PM is biologically equivalent to telling an adult to fall asleep at 6 PM — the brain's clock simply isn't ready. This shift reverses in the early twenties. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "You can train yourself to need less sleep" | Sleep need is biologically determined and not reducible through practice. Studies of people who claim to thrive on 4-5 hours of sleep show measurable cognitive impairment they've simply adapted to not noticing. The rare genetic variant (DEC2 mutation) that allows short sleep affects less than 1% of the population. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Catching up on sleep on weekends fixes everything" | Weekend sleep recovery partially repays sleep debt but doesn't restore all lost function. The irregular schedule creates 'social jet lag' that further disrupts the circadian clock. Research shows that chronic sleep restriction followed by recovery sleep does not fully restore cognitive performance, attention, or metabolic health. |